

IDRIS RAYAN CHAKIB

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Education

University of M'Hamed Bougara Boumerdes	Boumerdes, Algeria
Master of Science in Software Engineering and Data Processing	Sep 2024 - June 2026
Thesis Title: "AI Pipeline for Genetic Variant Analysis and Virtual Therapeutic Screening" (In partnership with Institut Pasteur d'Algérie). Academic Distinctions: Thésis Grade : 18/20 (Graduated with Highest Honors / Congratulations of the Jury). Startup Path (Decree 1275): Developed the project's core framework as an innovative startup. Startup Grade: 19/20 Awarded the Official National Startup Certificate. Key Coursework: Network and communication protocols, Cybersecurity, Distributed Systems, Software Architecture, Data Mining.	
University of M'Hamed Bougara Boumerdes	Boumerdes, Algeria
Bachelor of Science in Software Engineering and Information Systems	Sep 2021 - June 2024
Academic Standing: Final Grade : 10.98/20 Class Rank: 39 / 182 (Top 21.4% of the cohort / Equivalent to ECTS Grade B). Capstone Project Title: "Fraud Monitoring Web Solution with Machine Learning" (In partnership with Ooredoo Algeria). Project Distinction: Grade 16/20. Key Coursework: Object-Oriented Programming, Calculus, Linear Algebra, Probability & Statistics, Databases.	

Research Publication

- Benhenni O., Gaceb D., Touazi F., Rouibi B., Ismail A., Kebour W., **Idris R. C.**, Galleze A., Benidir M. "**SmartADN: An End-to-End Deep Learning Pipeline for Variant Pathogenicity Classification and Mutation-Aware Therapeutic Screening in the North African Context.**" Submitted to the IEEE International Symposium on iNnovative Informatics of Biskra (**IEEE ISNIB'26**), 2026. (Paper ID: 101 Currently Under Peer-Review / Forthcoming).
- Benhenni O., Gaceb D., Touazi F., Kebour W., **Idris R. C.**, Rouibi B., Ismail A. "**An End-to-End Approach for SNV Pathogenicity Prediction via Progressive Fine-Tuning of Nucleotide Transformer-v2.**" Published and presented at the 3rd National Conference on Applications of Artificial Intelligence (**A2I'26**), 2026. (Official Certificate of Participation Awarded as **Presenting Author**).

Professional Experience

Institut Pasteur d'Algérie	Algiers, Algeria
Data Processing & AI Research Intern	Feb 2026 – June 2026
- Co-designed SmartADN, an integrated pipeline coupling a sequence-only pathogenicity classifier with a mutation-aware therapeutic screening module. - Fine-tuned the Nucleotide Transformer-v2 genomic foundation model (500M parameters) using a layer-wise unfreezing strategy under a strict gene-split out-of-distribution (OOD) protocol. - Achieved a near-perfect AUC-ROC of 0.987 and 94% overall accuracy on entirely held-out clinically critical genes (BRCA1, BRCA2, TP53) without relying on population allele-frequency annotations. - Engineered the protein backbone integration for AEgis-GT, mapping conformational perturbations via the pointwise difference of ESM-2 embeddings ($\Delta e = eMUT - eWT$). - Implemented a bi-phasic training strategy using a linear zero-initialised adapter to successfully bridge biophysical Silver-Standard descriptors with experimental Gold-Standard affinity data.	
Ooredoo Algeria	Algiers, Algeria
Software Engineering Intern	Feb 2024 – June 2024
- Developed predictive machine learning models integrated into the enterprise telecom web infrastructure to track fraudulent patterns. - Analyzed large telecom datasets to improve data extraction pipelines, increasing tracking and software reliability.	

Projects

Multi-Robot Autonomous 2D Swarm Exploration
- Engineered a multi-robot swarm mapping system from scratch using a continuous Teacher-Student reinforcement learning strategy paired with the A* pathfinding algorithm. - Coded the entire Deep Q-Network (DQN) architecture, backpropagation loops, and neural layers using raw Python/NumPy.

Nexus: Decentralized P2P Encrypted Chat System

- Designed a robust peer-to-peer distributed framework featuring dynamic private chat rooms, logical isolation, and multi-threaded file transfer protocols.
- Implemented physical concurrency controls and data consensus layers using Lamport Logical Clocks, custom Byzantine-resilient voting systems, and multi-layered Fernet cryptographic encryption via Tkinter GUI.

Moogle: Natural Language Processing Search Engine

- Built a highly optimized localized text search engine integrating foundational Natural Language Processing (NLP) preprocessing pipelines.
- Developed custom algorithmic structures for tokenization, automated stop-word elimination, Porter-style stemming/unstemming, and contextual Bi-gram language modeling.

Leadership Experience

SmartADN Startup Project

Boumerdes, Algeria

Co-Founder & Lead AI Engineer

Sept 2025 – Present

- Selected for the prestigious Algerian National Student-Entrepreneur program under Ministerial Decree 1275, demonstrating academic and technical leadership.
- Architected and developed AEGIS-GT, a custom deep learning model designed for advanced genomic analysis and core startup technology.
- Successfully pitched the technical design and AI model architecture to academic committees, securing the official National Startup Certificate.

NASA International Space Apps Challenge

Global / Algiers Cohort

Team Leader & Core Developer

- Led a multidisciplinary team of student engineers and peers during the 48-hour global hackathon to address complex astronomical challenges.
- Co-designed and engineered a 3D interactive web-based Orrery application to simulate celestial body trajectories using open-source NASA telemetry data.
- Managed project workflows, delegated tasks among university peers under strict time constraints, and delivered the functional backend framework, awarded the **Galactic Problem Solver Certificate**

Additional Credentials & Activities

Professional Certifications

2025 – 2026

- **Machine Learning Specialization (DeepLearning.AI):** Certified core expertise in neural networks and AI system design.
- **Google Data Analytics Professional Certificate:** Certified proficiency in data cleaning, SQL analysis, and problem-solving.

Active Competitor, Codeforces Platform

2026 – Present

- Current Rating: 1166 (Actively progressing toward top-tier algorithmic milestones via timed challenges.)
- **Ranked 273rd out of 3,830 active global participants (Top 7.1%)** in the ICPC Online Challenge 1 (powered by Huawei); engineered a custom scheduling heuristic to optimize concurrent request routing for Edge-Cloud Collaborative LLM Inference loops.

Technical Skills

- **Languages:** Java, Python, C/C++, SQL, JavaScript, HTML/CSS.
- **Frameworks:** Next, Express, FastAPI, Flask, Pytorch, Keras, Scikit-learn, Tensorflow, Hugging Face (Transformers).
- **Developer Tools:** Git, Docker, Node.js, VS Code, Visual Studio, Pycharm, IntelliJ, Eclipse.
- **Libraries:** React, pandas, Numpy, Matplotlib, seaborn.